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(ITCS4315) Data Science

Final Project Report – Data Science Applications

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Introduction

In my project work, I applied the complete cycle of Data Science. Data Science involves collecting data, cleaning and preparing it, exploring and visualizing patterns, and finally applying machine learning models to generate insights. The general workflow I followed included:

1. Collecting raw datasets from sources such as CSV files and web scraping.
2. Preprocessing the data by handling missing values, removing duplicates, and converting data types.
3. Exploring and visualizing the data using histograms, boxplots, and scatterplots to understand distributions and detect outliers.
4. Applying machine learning models for classification, regression, and clustering depending on the problem.
5. Evaluating the models using accuracy, error, and variance explained.

In my own work, I also used web scraping to collect product data from online sources. This was then cleaned and analyzed, showing how I moved from raw data acquisition to actionable insights.

Question 1 – Student Impact Dataset

Dataset Description:

- I worked with a dataset of 50,000 rows and 18 columns.
- The dataset included features such as PreSemesterGPA, PostSemesterGPA, WeeklyGenAIHours, TraditionalStudyHours, SkillRetentionScore, and AnxietyLevelDuring_Exams.
- The target variables were BurnoutRiskLevel (categorical) and PostSemesterGPA (numeric).

Preprocessing Steps:

1. I checked the dataset for missing values. Small gaps were found in PreSemesterGPA, PrimaryUseCase, and Institutional_Policy.
2. I filled missing values with “not available” to keep the dataset consistent.
3. I removed duplicate rows to avoid bias.
4. I converted GPA columns to numeric values and dropped invalid entries.
5. After these steps, the dataset was clean and ready for analysis.

Visualization:

- I created a histogram and boxplot of PreSemesterGPA to show the distribution of GPA values. Most students were between 2.0 and 4.0, with some outliers.
- I plotted a confusion matrix to evaluate classification results for burnout risk.
- I created a scatter plot comparing predicted versus actual GPA values, which showed close alignment.

Machine Learning Models:

1. Decision Tree Classifier

- Target: BurnoutRiskLevel (High, Medium, Low).
- Accuracy: about 51%.
- Precision and recall varied across classes, with High performing better and Medium weaker.
- Reason for choice: Decision Trees are interpretable and suitable for categorical targets.
- Limitation: Moderate accuracy due to the complexity of psychological and behavioral data.

2. Linear Regression

- Target: PostSemesterGPA.
- Features: PreSemesterGPA, study hours, AI usage, skill retention, and anxiety level.
- R^2 Score: 0.87 – strong explanatory power.
- Mean Squared Error: 0.03 – low error.
- Reason for choice: Regression is ideal for continuous numeric prediction.
- Result: Model successfully predicted GPA outcomes with high reliability.

Results and Conclusion – Question 1:

- In my project, classification achieved moderate success with 51% accuracy. It was useful for identifying burnout risk but not highly precise.
- Regression achieved strong success with $R^2 = 0.87$, making it a reliable predictor of GPA outcomes.
- I used classification because the burnout risk was categorical, and regression because GPA was continuous.
- My final insight was that regression provided the most actionable results, while classification highlighted the challenge of modeling complex human behaviors.

Question 2 – AyatStore Products Dataset

Dataset Description :

In the second part of my project, I worked with a dataset of 1,094 rows and 9 columns. The dataset contained product information such as category, name, price, oldprice, discount, producturl, image_url, and description. My goal was to clean the raw product data, prepare it for analysis, and then apply clustering to group products by price.

Preprocessing Steps:

I began by reading the raw CSV file and splitting the columns correctly. I renamed the columns to make them clear and added a new description column with a default value of “not available.”

- I cleaned the price and old_price columns by extracting numeric values and converting them into floats.
 - I handled missing values by filling them with “not available.”
 - I dropped rows that had no product name or price, and I removed duplicates.
- After these steps, the dataset was consistent and ready for analysis.

Visualization:

I first visualized the raw price values using histograms and boxplots, which showed invalid string values. After cleaning, I plotted the price distribution again and saw a logical spread, with most products priced between 3 and 10, and some outliers up to around 35.

I then normalized the price column using MinMaxScaler, creating a new column called price_norm with values between 0 and 1. I visualized the normalized prices with histograms and boxplots, confirming that all values were scaled correctly.

Machine Learning Model:

I applied KMeans clustering to group the products into three clusters based on their price values.

- Cluster 0 contained higher-priced products (around 10.0 and above).
- Cluster 2 contained medium-priced products (around 3.0–6.0).
- Cluster 1 included other ranges.

I visualized the clusters using a scatterplot of price versus old_price, with colors showing the cluster assignments. I also plotted a histogram of product distribution by cluster, which confirmed that the clustering model successfully grouped similar products together.

Results and Conclusion – Question 2:

In my project, the clustering model grouped products into three meaningful categories based on price. This allowed me to identify high, medium, and lower price ranges. The scatterplot and histogram confirmed that the clusters were logical and consistent. I chose KMeans because it is effective for unsupervised grouping when labels are not available, and it provided clear insights into product segmentation.

Final Course Conclusion :

Through this course and my final project work, I learned how to apply the complete cycle of Data Science: collecting and scraping data, cleaning and preprocessing, visualizing distributions, and applying machine learning models. I practiced classification, regression, and clustering, and I understood their strengths and limitations.

- Classification helped me explore categorical outcomes such as burnout risk, though accuracy was moderate.
- Regression gave me strong predictive power for GPA outcomes, showing how continuous variables can be modeled effectively.
- Clustering allowed me to group products logically without labels, demonstrating the value of unsupervised learning.

Overall, I gained practical experience in handling real datasets, preparing them for analysis, and applying different machine learning techniques. This project improved my technical skills and gave me confidence in using Data Science methods to solve diverse problems.

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